Surprise Test- Solution Sketch

1.1

P(X=a) = 0.32 + 0.08 = 0.4
P(X=b) = 0.48 + 0.12 = 0.6

P(Y=a) = 0.32 + 0.48 = 0.8
P(Y=b) = 0.08 + 0.12 = 0.2

P(X=a,Y=a)=0.32, P(X=a) * P(Y=a) = 0.4 * 0.8= 0.32, ...works
P(X=a,Y=b)= 0.08, P(X=a) * P(Y=b) = 0.4 * 0.2 .....works
Similarly,
0.48 = P(X=b) * P(Y=a) = 0.6 * 0.8 works
0.12 = P(X=b) * P(Y=b) = 0.6 * 0.2 works

-> Independently distributed

1.2

H(X) = -(0.4 * (-1.322) + 0.6 * (-0.737)) = 0.971
H(Y) = -(0.8 * (-0.322) + 0.2 * (-2.322)) = 0.722
H(X, Y) = -(0.32 * (-1.644) + 0.08 * (-3.644) + 0.48 * (-1.059) + 0.12 * (-3.059)) = 1.693

H(X|Y) = H(X, Y) - H(Y) = 1.693 - 0.722 = 0.971
H(Y|X) = H(X, Y) - H(X) = 1.693 - 0.971 = 0.722

1.3

D(X||Y) = 0.4 * (-1.322 - (-0.322)) + 0.6 * (-0.737 - (-2.322))
        = 0.551

2. a) Using the given assumptions, we can easily define $P(N = \texttt{name})$ for newspaper text as well as for scientific text:

$$\begin{aligned} P_{\text{Newspaper-Text}}(N = \texttt{name}) &= 0.05 \\ P_{\text{Scientific-Text}}(N = \texttt{name}) &= 0.01 \end{aligned}$$

Using *Bayes Theorem*, we can calculate the probability

$P(N = \texttt{name}|C = \texttt{yes})$ as follows:

$$P(N = \texttt{name}|C = \texttt{yes}) = \frac{P(C = \texttt{yes}|N = \texttt{name})P(N = \texttt{name})}{P(C = \texttt{yes})}$$

with

$$\begin{aligned} P(C = \texttt{yes}) = \; & P(C = \texttt{yes}|N = \texttt{name})P(N = \texttt{name}) + \\ & P(C = \texttt{yes}|N = \texttt{not_name})P(N = \texttt{not_name}) \end{aligned}$$

yielding the following results:

$$\begin{aligned} P_{\text{Newspaper-Text}} &= \frac{0.9 \cdot 0.05}{0.235} = \frac{9}{47} = 0.1915 \\ P_{\text{Scientific-Text}} &= \frac{0.9 \cdot 0.01}{0.207} = \frac{1}{25} = 0.043478261 \end{aligned}$$

2. b) How low must the false positive rate $P(C = \texttt{yes}|N = \texttt{not_name})$ get so that this probility goes up to 50% for both kinds of text?

$$
\begin{aligned}
& P(N = \texttt{name}|C = \texttt{yes}) && \geq && 0.5 \\
\Leftrightarrow \quad & \frac{P(C = \texttt{yes}|N = \texttt{name})P(N = \texttt{name})}{P(C = \texttt{yes})} && \geq && 0.5 \\
\Leftrightarrow \quad & && \ldots \\
\Leftrightarrow \quad & P(C = \texttt{yes}|N = \texttt{not_name}) && \leq && \frac{P(C = \texttt{yes}|N = \texttt{name})P(N = \texttt{name})}{P(N = \texttt{not_name})}
\end{aligned}
$$

Now, we can calculate $P(C = \texttt{yes}|N = \texttt{not_name})$ for both kinds of text resulting in:

$$
\begin{aligned}
P_{\text{Newspaper-Text}}(C = \texttt{yes}|N = \texttt{not_name}) &= \frac{0.9 \cdot 0.05}{0.95} = 0.0474 \\
P_{\text{Scientific-Text}}(C = \texttt{yes}|N = \texttt{not_name}) &= \frac{0.9 \cdot 0.01}{0.99} = 0.0091
\end{aligned}
$$

Thus, in order to classify at least 50% of the names correctly, the false positive rate has to be below 4.74% for newspaper text and below 0.91% for scientific text.